Engineering Pulse

Multi-agent PR review with an Engineering Manager decision layer

Engineering Pulse uses AI to surface engineering evidence, but management judgment stays human.



PR-level question

What is wrong with this change?



Manager-level question

What pattern is forming across the team, and what should I do about it?



Most AI review tools stop at the first question.
This project is built around the second.

Velocity without context is not productivity.

Throughput

\+ Complexity

\+ Review quality

\+ Severity

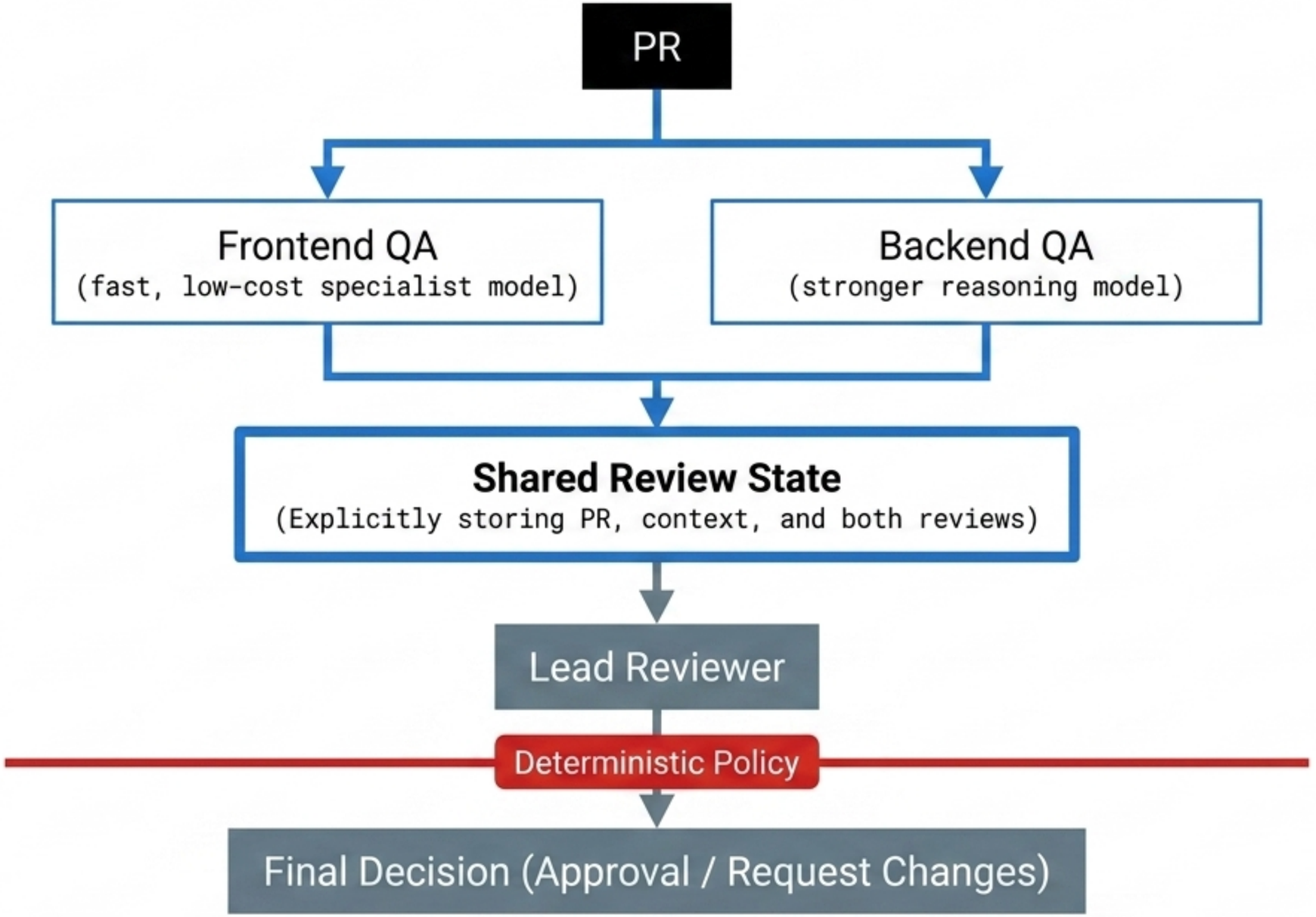
\+ Trend

= Better management signal

The Archetype: Nadine

Complexity: 5 (Highest)	First-pass rate: 100%	Defect density: Low

Nadine exemplifies this philosophy: her lower raw PR volume is offset by the highest assignment complexity and a 100% first-pass approval rate.



AI handles specialist reasoning. Code handles policy guarantees.

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Why deterministic guardrails matter



Blocking rules

High and critical findings cannot be casually approved. Severity takes precedence over reviewer count.



Risk calculation

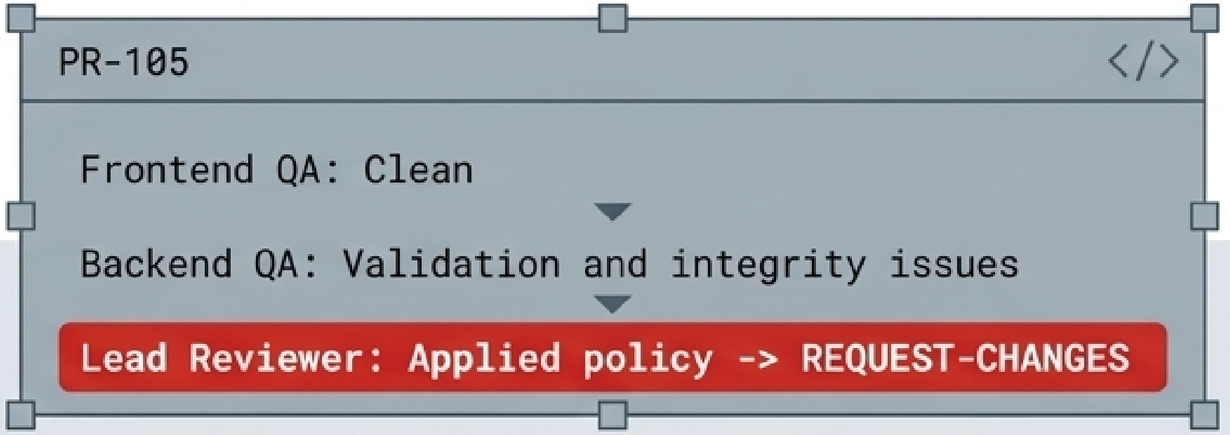
The model identifies and explains risk; deterministic code scores it.



Evidence requirement

Formal findings must be supported by the supplied PR. Unsupported possibilities become verification notes.

I want AI flexibility where reasoning helps and deterministic control where consistency matters.



Team at a glance

Dataset Scope:
8 seeded PRs

Risk Profile:
4 high-risk findings



Overview screenshot

Team Performance:
63% first-pass approval

Pattern Recognition:
11 structured findings

Note: These numbers are generated from a deterministic mock assessment dataset for reproducible evaluation. They are not claims about a real production engineering organization.

Workforce diagnostics: context changes the story

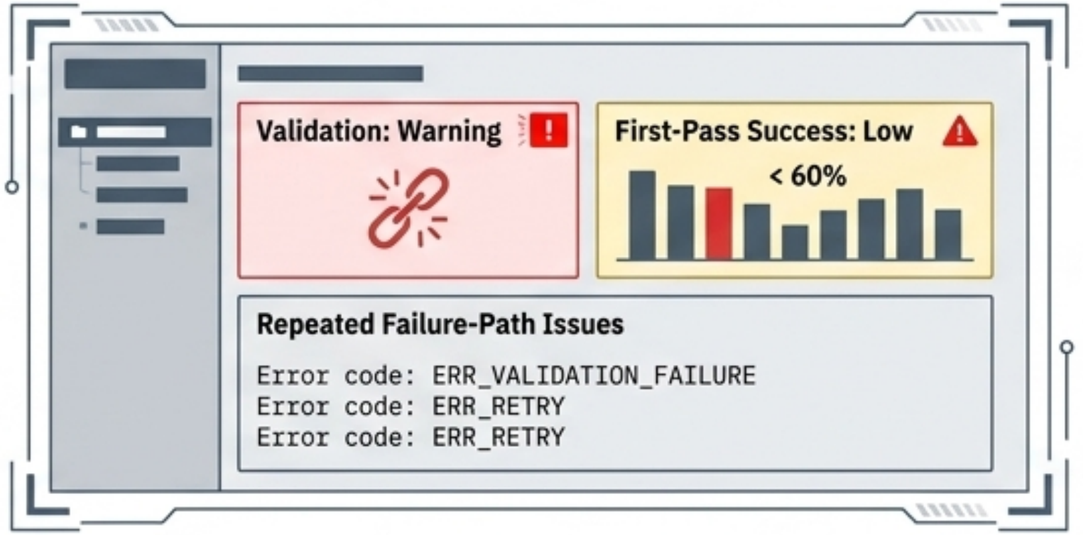
Nadine



- Highest assignment complexity
- 100% first-pass approval
- Low defect density

Conclusion: Lower raw PR volume is not weak performance. High-complexity ownership explains the output rate.

Luka



- Repeated validation and failure-path issues
- Lower first-pass success

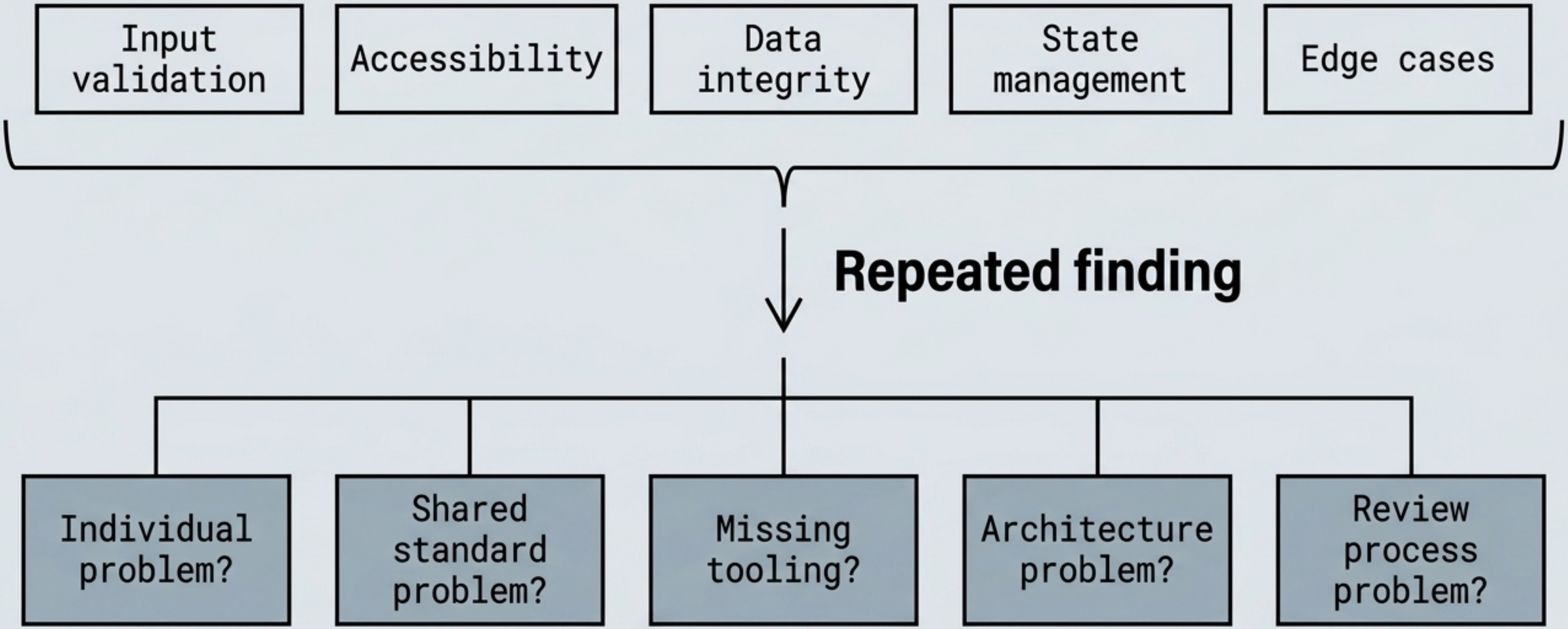
Conclusion: Specific coaching signal, not a general performance verdict.

Coaching should be a testable hypothesis

	Pattern	Action	Observable Next Step
Luka	Repeated validation and failure-path misses	Targeted pairing + Add lightweight edge-case checklist	Observe the next 3 comparable PRs to see if the specific pattern improves
Theo	Initial inconsistency, followed by an improving quality trend	Gradually increase ownership + Reinforce successful habits	Evaluate sustained quality on progressively more complex work

Intervention should produce an observable hypothesis,
not just a vague instruction to “do better”.

Team bottlenecks require system thinking



The manager's job is to determine the level at which the problem should be solved

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What I learned building it

Hard Lessons

Demo-Data Isolation

Live models must not see seeded expected answers. Demo findings must be strictly stripped before live evaluation.

AI Constraint

Model judgment needs deterministic policy boundaries. A model can reason correctly but return an inconsistent numeric score.

Evidence Boundaries

Unsupported possibilities should not become formal findings. They belong strictly in verification notes.

Infrastructure Reality

Provider availability must be treated as a system dependency. Model routing must remain configurable.

PR analytics can surface a conversation. They cannot make a people decision.

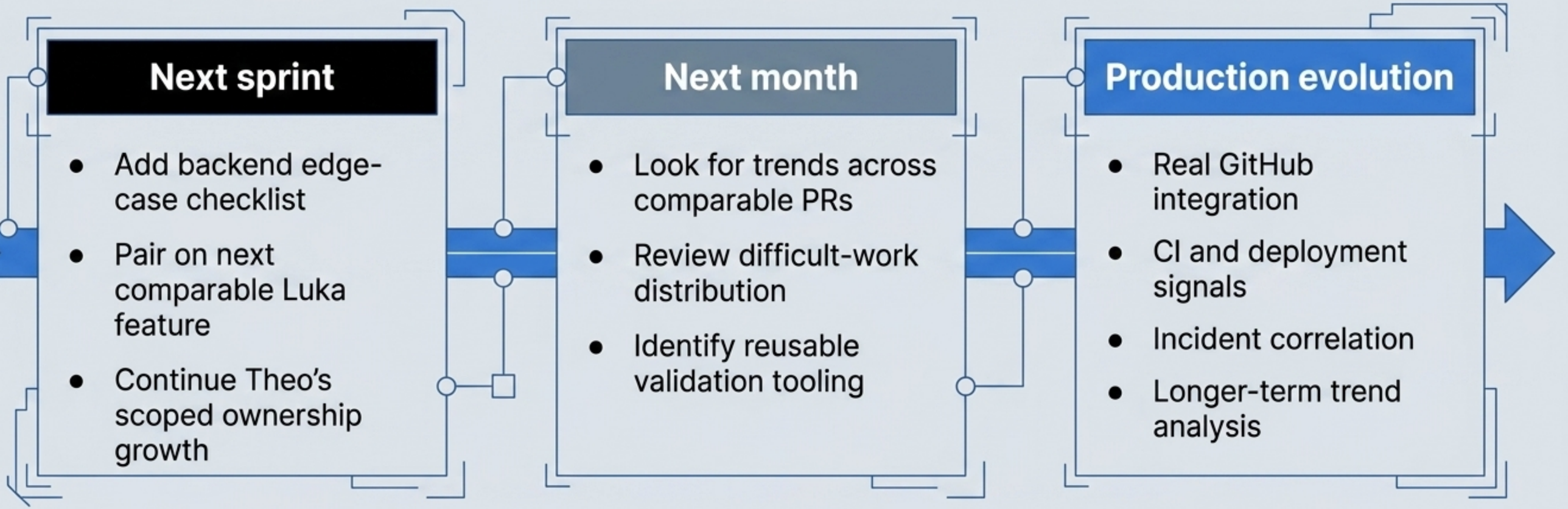
What the system does NOT infer:

~~Burnout~~ ~~Motivation~~ ~~Engagement~~ ~~Promotion readiness~~ ~~Termination readiness~~

Responsible management requires context from the engineer, the team, and the actual work.

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What I would do next as Engineering Manager



AI surfaces evidence. Managers remain accountable for judgment.